

Predicting Customer Booking Behaviour

Analytical summary for the Customer Insights team

MODEL PERFORMANCE

Test ROC-AUC

0.774

on held-out data

5-Fold CV AUC

0.762

± 0.005 — stable across folds

Model

Random Forest

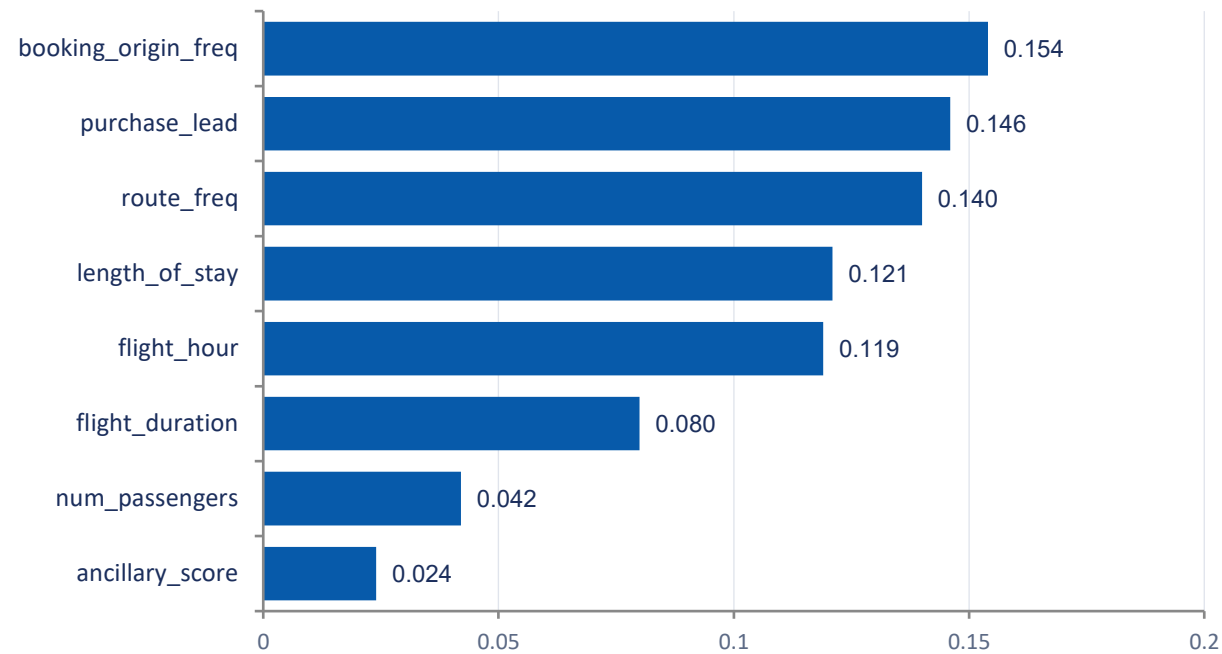
pruned to top 8 features

Class Balance

85 / 15

non-bookers vs. bookers

TOP DRIVERS OF BOOKING PROBABILITY



KEY FINDING

Geography and timing drive booking intent.

The top 6 features are all origin, route, or timing signals. Model performance clusters at ~0.77 AUC across four algorithms (LogReg, RF, GB, XGB), indicating the ceiling is set by the input data rather than model choice.